

The mantle, CO₂ and the giant Aptian chemogenic lacustrine carbonate factory of the South Atlantic: Some carbonates are made, not born

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ABSTRACT

During the Aptian (Cretaceous), in what is now the South Atlantic, the largest chemogenic (abiotic) carbonate factory so far identified in the Phanerozoic geological record developed as a vast hyper-alkaline lake system. This covered at least 330 000 km², producing carbonates, locally over 500 m thick, in what are now the offshore Santos and Campos basins (Brazil), and Kwanza Basin (Angola). Current evidence supports the view that almost all of this carbonate was chemogenic in origin, precipitated from hyper-alkaline, shallow lake waters, probably by evaporation. This unit, best documented from offshore Brazil and known as the Barra Velha Formation (Santos Basin) and the Macabu Formation (Campos Basin), consists of just two basic carbonate components, millimetre to centimetre sized crystal shrubs and spherulites. These are commonly *in situ* but can also be reworked into a range of detrital facies. Demonstrable microbialites are generally rare. These carbonates are associated with Mg silicates (as clays) which had a profound influence not only on the textural development of the *in situ* carbonates, but also on their diagenesis. The dissolution of the clays produced much of the porosity in these limestones, which are the hosts for multi-billion barrel oil fields. The source of the carbonate was most likely from metasomatic alteration of mafic rocks, such as continental flood basalts related to Atlantic opening, with some contribution from much older continental basement. Clear evidence that serpentinization of possible exhumed mantle is lacking but mantle CO₂ is likely to have been a critical factor in determining the composition of the fluids from which the carbonates formed and the high alkalinities of the lake waters.

Keywords Carbonate factory, CO₂, Cretaceous, lacustrine, mantle, South Atlantic.

INTRODUCTION

The flux of Ca and Mg and dissolved inorganic carbon to the oceans, and ultimately to the marine carbonate systems, is controlled by the interaction of tectonics and atmospheric CO₂ (Arvidson *et al.*, 2013, and references therein). In addition, it is not just the flux that is controlled by this interaction but also the mineralogy of chemogenic

and biogenic carbonates, via Mg/Ca, alkalinity and temperature, and even extinction events impacting on aragonite/calcite (Zhuravlev & Wood, 2009; Balthasar & Cusack, 2015). The role of tectonics involves those large-scale processes affecting rates of continental crustal weathering but also seawater–basalt reactions at spreading ridges, and also serpentinization of exhumed mantle. A view has been reached that

atmospheric CO₂ levels are a critical factor in release of the required ions for carbonate accumulation, including high rates of volcanic degassing of CO₂, which have been proposed as a critical factor in the high neritic carbonate preservation and production rates which characterize the mid-Cretaceous (Pohl *et al.*, 2020).

Existing classifications of carbonate systems and factories focus mainly on the biotic, including microbial, marine examples, emphasizing ecological controls (Schlager, 2003; Pomar & Hallock, 2008; Michel *et al.*, 2018; Laugié *et al.*, 2019; Pomar, 2020). During the Archean to Palaeoproterozoic, however, extensive precipitation of chemogenic carbonates in marine settings took place (Grotzinger & James, 2000). Less emphasis has been placed on lacustrine carbonates, especially the nature of their carbonate factories and sediment budgets (see review in Pelechaty *et al.*, 2013). While the supply of their elemental components to non-marine carbonate systems, especially lakes, is also largely influenced by tectonics and atmospheric CO₂, the nature of non-marine carbonate systems also reflects complex local hydrological and hydrogeological controls (Fig. 1).

A critical influence on the chemical products of lacustrine systems is the nature of the

catchment geology (Hardie & Eugster, 1970). In endorheic (closed) lake basins the precipitates are influenced by the initial ion concentrations of the inflow waters and the types and amounts of minerals that precipitate, via chemical divides (Hardie & Eugster, 1970). Although the diversity of precipitates in such lakes is well understood by reference to the influence of these divides, quite specific catchment geologies can produce unusual systems (Cerling, 1994). Such considerations are especially important as many large lake basins, past and present, are associated with rifting (e.g. Owen *et al.*, 2019).

The supply of water to such lakes comes from run-off, direct precipitation and groundwater recharge. For example, the large, evaporitic, carbonate producing Great Salt Lake rift basin in Utah (USA), receives 66% of its water from run-off related to snowmelt in nearby mountains, with 31% from direct precipitation and only 3% from groundwater. However, in many balance-filled and underfilled rift basins groundwater-fed springs (shallow meteoric or hydrothermal) are common (Owen *et al.*, 2018). An extreme example of a spring-filled lake is Nasikie Engida, to the north-west of Lake Magadi (Kenya) where an estimated 80 to 90% of inflow is from hot springs (Owen *et al.*, 2018). Hydrothermal waters

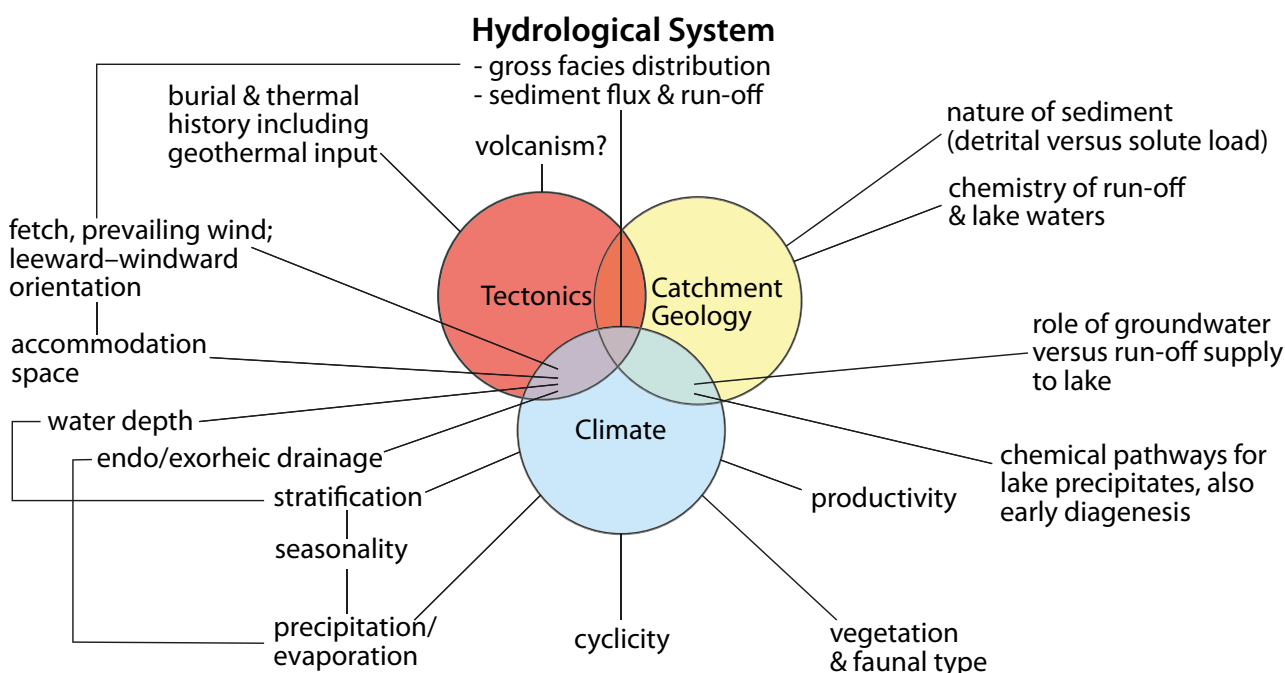


Fig. 1. Simplified consideration of the major controls and interactions and how these affect depositional systems in lake basins (modified from Wright & Rodriguez, 2018).

associated with magmatism are commonly alkaline and CO₂ charged (e.g. Lowenstein *et al.*, 2016). The balance between the solutes coming from run-off versus groundwater will also be affected by changes in evaporation–precipitation and climate (e.g. Mercedes-Martín *et al.*, 2019).

The aim of this study is to synthesize current knowledge on the Aptian giant lacustrine carbonate factory of the central South Atlantic, and propose that it reflects the direct influence of mantle-related CO₂ on basement geology and volcanic catchments, resulting in a vast chemogenic carbonate factory, which seems to have had more in common with the Archean world than any other carbonate system, marine or non-marine, in the Phanerozoic.

GEOLOGICAL SETTING

During the early Cretaceous, continental break-up associated with rifting created extensive lacustrine basins in what are now the South Atlantic marginal basins of the eastern seaboard of South America, and West Africa (see review in Sabato Ceraldi & Green, 2017; Kukla *et al.*, 2018). This established the related Santos–Namibe basin system (Fig. 2A), and the Espírito Santo and Campos basins and their conjugates (Strozyk *et al.*, 2017).

The deposits of these lakes constitute the Guaratiba Group of Santos Basin, the Lagoa Feia Group of Campos Basin; Fig. 2B) and the Toca-Chela of West Africa (Saller *et al.*, 2016). Initially, during the earlier rifting stage, alluvial (Piçarras Formation) and later lacustrine deposits (Itapema Formation; Fig. 2B) were laid down in Santos Basin. These lakes were relatively deep, in which extensive mollusc coquinas developed (e.g. Barnett *et al.*, 2020). During the later rift stage and the overlying sag phase (Sabato Ceraldi & Green, 2017) the lakes became highly restricted and alkaline (Wright & Barnett, 2015). This unit is known as the Barra Velha Formation (BVF) in Santos Basin and is the focus of this study. Its equivalent in Campos Basin is the Macabu Formation, and elsewhere in Brazil the Codo Formation, and the Chela in Kwanza Basin, Angola. The BVF ranges in thickness from >550 m to as little as <55 m or is missing on fault block highs (Wright & Barnett, 2015).

The basement rocks of Santos Basin below the Cretaceous consist of the crystalline basement and younger basalts. The former are the

equivalent of outcrops in the Serra do Mar area in São Paulo state, which are mainly Precambrian granitoids and gneisses (Moreira *et al.*, 2007). These are unconformably overlain by tholeiitic (low Na) basalts of the Camboriú Formation (Fig. 2B) which are likely Hauterivian–Aptian in age (Pietzsch *et al.*, 2018), and are also intercalated with the BVF in Santos Basin and its equivalent in Campos Basin (Szatmari & Milani, 2016). On structural highs the BVF rests directly on basalts. The rifting that produced the lake basins may have resulted in their floors being possibly as much as 700 m below that of the contemporaneous sea-level (Reston, 2010).

The BVF (Alagoas Stage) is separated from the Itapema Formation (Jiquiá Stage) by the Pre-Alagoas unconformity. Internally the BVF is divisible by the Intra-Alagoas unconformity, which is regarded as separating the informally defined Rift from Sag units. The BVF (Fig. 2B) is Aptian in age, within the Alagoas Stage in Brazil (Moreira *et al.*, 2007; Winter *et al.*, 2007; Petersohn & Abelha, 2013) and, based on this, was deposited between 123 to 115 Ma (e.g. Wright & Barnett, 2020). Tedeschi *et al.* (2017) have suggested that the uppermost part of the BVF coincides with the onset of the Oceanic Anoxic Event 1a (OAE 1a), so before *ca* 120 Ma or below *ca* 125 Ma, according to different calibrations. Pietzsch *et al.* (2018, fig. 2) regard the age of the BVF as between 123 to 112 Ma. They later (Pietzsch *et al.*, 2020), by comparisons of the C isotope curves from Well S10 from Santos Basin and the Cismon core (Italy), imply deposition of the BVF as likely 125 to 128 Ma and prior to the onset of OAE 1a (*ca* 120 to 119 Ma).

The BVF is overlain by the Ariri Formation, a package of over 1 km of marine-derived evaporites representing the Cretaceous ocean seeping and pouring into the basin (Davison *et al.*, 2012). The climate during the Aptian and Albian in this southern part of the central South Atlantic is regarded as having been arid (Lentini *et al.*, 2010; Chaboureaud *et al.*, 2012).

The facies assemblage which comprises the BVF is recorded from the Campos (Herlinger *et al.*, 2017) and Kwanza basins (Saller *et al.*, 2016; Sabato Ceraldi & Green, 2017) and there is even a minor development in the Codó Formation of the Parnaíba Basin in north-eastern Brazil (Bahniuk *et al.*, 2015). An estimate based on the extent of the BVF facies assemblage in the three basins (Santos, Campos and Kwanza) suggests that the carbonate system was potentially active over at least 330 000 km².

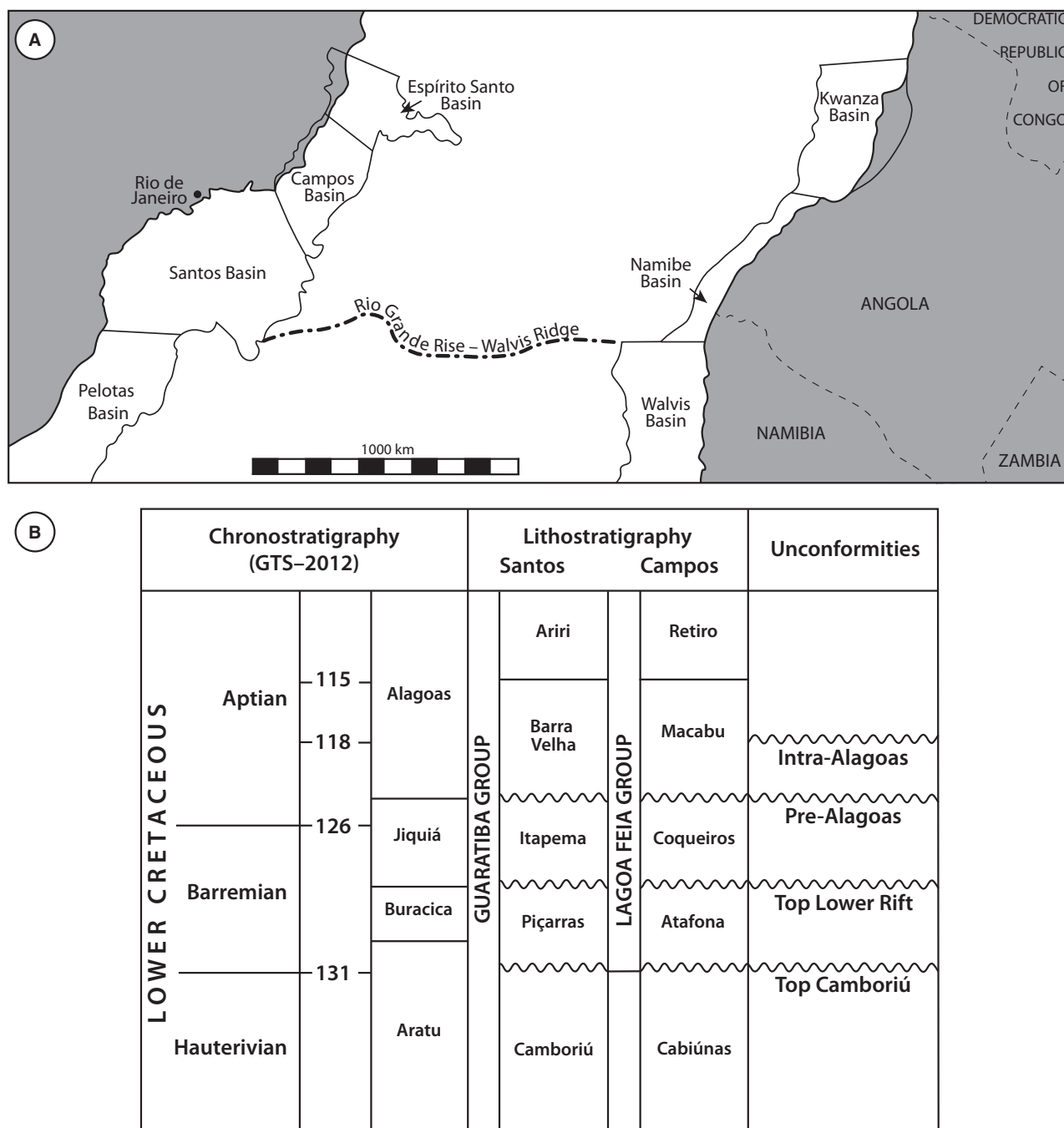


Fig. 2. (A) Location map of Santos Basin and related basins. (B) Lower Cretaceous stratigraphic chart for the Santos and Campos basins (after Moreira *et al.*, 2007; Winter *et al.*, 2007).

While the BVF has been referred to informally as the ‘Microbialites’ (e.g. Kattah, 2017), Wright & Barnett (2015, 2020) have, for Santos Basin, provided data on the paucity of evidence for microbialites based on the examination of 3502 thin sections from 19 wells and 10 oil fields and drilled structures. Microbialite facies from units

immediately below the Ariri salt and its equivalents, in what appear to be more marginal marine depositional systems, have been provided by Muniz & Bosence (2015) from the Macabu Formation of the southern Campos Basin, as well as from the Muribeca Formation (Aptian) of the Carmópolis field, Sergipe–Alagoas Basin by De

Araujo *et al.* (2012), and the Codó Formation of the Parnaíba Basin by Bahniuk *et al.* (2015). Hunt *et al.* (2019) report biogenic microbialites and breccias deposited in a range of saline to alkaline lake waters from Campos Basin, represented by diverse microbial and related features not seen in the BVF and its equivalents.

PLATFORM SETTING OF THE BARRA VELHA FORMATION

The gross depositional setting for the carbonates of the BVF is the subject of debate. Based on seismic data, a number of authors have interpreted the facies characterizing the BVF and its equivalents as representing deposition on isolated carbonate platforms within relatively deep lacustrine basins, with elevations of the platforms of hundreds of metres (der Mann & Rigg, 2012; Buckley *et al.*, 2015; Saller *et al.*, 2016; Sabato Ceraldi & Green, 2017; Kattah, 2017; Vieira de Luc *et al.*, 2017; Ferreira *et al.*, 2019). As a result, some authors have proposed distributions of key lithofacies based on the view that such facies would be differentiated on shallow platform tops (Saller *et al.*, 2016; Sabato Ceraldi & Green, 2017) and even populating reservoir models on that basis (de Paula *et al.*, 2017). The evidence used to identify the existence of isolated high relief platforms has come from their current structural positions on fault blocks and from clinoformal geometries implying depositional relief. While lacustrine carbonate systems are capable of producing small-scale platform-like depositional geometries (Platt & Wright, 1991), such as from the Eocene of the Bohai Bay Basin, China (Liu *et al.*, 2020), the relief on such platforms was decametre in scale at best and stratigraphically thick successions are not recorded. Shallow water lacustrine carbonates are known to have developed on pre-existing structural relief and hence were adjacent to deeper lake waters such as in the Toca of the Congo Basin, West Africa (for example see Wright, 2012).

The three basins in which the distinctive facies characterizing the BVF developed are structurally complex with many studies highlighting potential effects that can make interpreting seismic features and tectonic history far from straightforward. These include the extension discrepancy and polyphase faulting discussed by Reston (2009) and McDermott & Reston (2015). The effects of outer margin

collapse (Pindell *et al.*, 2014, 2018) during the late Albian are another consideration and could have triggered deformation affecting the BVF and produced relief on the top of that unit, easily misinterpreted for depositional (topographic) relief. Alves *et al.* (2017) record, from Santos Basin, the effects of inversion when normal faults, formed during the syn-rift stage were later reactivated as low and high-gradient reverse faults and pop-up anticlines. On some seismic lines, inversion structures, related to normal faults, intersect the base salt, and significant erosion took place locally at the top reservoir level, above the area where inversion is recorded. Karner & Gamboa (2007) record faulting and erosion having affected the BVF prior to salt deposition. Davison *et al.* (2012) record many small normal fault offsets at the base salt level, a small number with offsets of more than 1 km. Later rapid salt deposition and movement (gravitational gliding) caused significant fault reactivation and enhanced structural relief along the base of the salt (Davison *et al.*, 2012; Alves *et al.*, 2017). An additional issue is that volcanics can produce a range of seismic features readily misinterpreted as carbonate in origin such as clinoforms and mounds (Wright & Barnett, 2017).

Thus, identifying depositional topography in the BVF and its equivalents requires careful screening. One approach to this is to apply the methodology and criteria developed for identifying marine isolated carbonate build-ups by Burgess *et al.* (2013). Although very little is known about seismic geometries associated with lacustrine carbonates, either modern day or in the stratigraphic record, if light dependent microbialites were the major sediment producers for the BVF facies, the approach by Burgess *et al.* (2013) should be applicable, as their focus was largely on the behaviour of tropical, phototrophic carbonate factories. Those published studies describing isolated platforms in the BVF and its equivalents have not used any such approach in their published interpretations. Wright & Rodriguez (2018) provide a discussion of these issues in relation to the BVF in Santos Basin.

Wright & Barnett (2017) stressed that no modern lacustrine carbonate systems produce large differentiated platforms, but that there are examples of isolated, sub-lacustrine and subaerial mounds, the former at least of seismic-scale (see below). Those authors also discussed issues relating to the apparent presence of clinoforms in Santos Basin, which might be used as

evidence of high relief carbonate margins. The main evidence for the absence of any significant relief during deposition of the BVF came from a correlation of shallow water facies for 180 km west to east across Santos Basin using the package of carbonates at the top of the BVF with a distinct gamma log pattern known in the industry as Lula's Fingers. This package is 20.8 to 28.5 m thick (mean = 24.3 m), and shows nine gamma-defined cycles (mean thickness 2.7 m), which comprise one or more shallowing-upward cyclothems. These are well sampled by core and side wall core (SWC) and their facies are very similar in all wells, including a range of unequivocally shallow water facies (for example, microbial laminites and stromatolites; Wright & Barnett, 2017; Pietzsch *et al.*, 2020). This suggests that as these cyclothems were deposited at the same shallow water depths, and because such conditions prevailed across the basin, by inference, there were no deeper off-platform areas. At present some of these sections are separated by >1 km vertical relief indicating significant post-depositional deformation. An important issue here is that almost all of the data in the public domain about the BVF and its equivalents comes from wells positioned on what are now structural highs. This might suggest that these data sets are skewed to shallow water settings, as the correlations by Wright & Barnett (2017) show, and no data is available from locations off the highs.

Except for the facies in the Lula's Fingers interval, the BVF and its equivalents contain few indicators for water depth other than wave influenced grainstones–rudstones, and associated facies described below. The close stratigraphic association of the more common BVF facies with these grainstones–rudstones suggests that water depths were indeed shallow. Further support comes from C and O isotope studies, using the approach initially developed for East African lakes by Talbot (1990), that indicate the BVF carbonates were deposited in lakes with a large area:depth, that is, shallow (Sabato Ceraldi & Green, 2017; Wright & Barnett, 2017; Pietzsch *et al.*, 2018, 2020) and, as discussed below, strongly affected by evaporation. If the BVF facies represent shallow interiors of isolated carbonate platforms adjacent to large and deep water bodies, these interiors were isolated, hydrologically, from the deeper lake system, if that ever existed during BVF times. If the BVF represents deposition in a shallow lake, from C and O stable isotopes and $^{87}\text{Sr}/^{86}\text{Sr}$ data from the Campos and Santos basins (e.g. Pietzsch

et al., 2018, 2020), Lima *et al.* (2020) propose that either a single vast lake existed or that the lakes were hydrogeologically connected.

There are what appear to be high relief isolated mounds in Santos Basin described by Buckley *et al.* (2015) and Barnett *et al.* (2018) with seismic relief of 400 m. The mounds are fractured and karstified and are associated with faults, forming over fault-related antecedent highs. The cores of these features consist of *in situ* shrubs (see below) with surrounding deposits of grainstones, rudstones and, less commonly, packstones and floatstones composed of reworked shrubs, with platy, angular and well-rounded intraclasts mainly comprising centimetre to decimetre-scale fragments of *in situ* shrub cementstones. Grain-size is mainly coarse-grained to very coarse-grained to pebble-grade, with poor to moderate sorting with sub-rounded to rounded grains. Packages of beds commonly show elevated dips of *ca* 20 to 40°, and show normal grading but mainly lack internal stratification. Barnett *et al.* (2018) interpreted these reworked carbonates as colluvial wedges resulting from the subaerial disaggregation of mounds. Souza *et al.* (2018), referring to such occurrences, cite formation in a low-temperature hydrothermal and evaporitic environment, linked to ascending Ca–Mg–Si rich fluids emerging in subaerial and sub-lacustrine springs. There are some issues arising from the occurrence of these features. Firstly, they do not appear to show the range of textures and geometries associated with subaerial tufas or travertines, and their scale is unlike any such features known today. Thick subaerial travertine deposits are known from, for example, the Ballik area, Denizli Basin in West Anatolia, Turkey (Claes *et al.*, 2015), but consist of stacked travertine wedge units with diverse internal facies now well documented from many parts of the world (e.g. Della Porta *et al.*, 2017a,b).

High relief, isolated lacustrine carbonate build-ups are known from modern lakes such as Pyramid Lake in Nevada, which can be >100 m high, and are now accessible because of lake level falls, but formed sub-aqueously, typically around vents discharging Ca-rich waters into carbonate-rich lakes. These features have complex internal differentiation (Della Porta, 2015; DeMott *et al.*, 2018), seemingly unlike those in Santos Basin which appear in part strongly layered from published seismic images. The features described by Barnett *et al.* (2018) were also located near faults, strengthening a link to

sub-lacustrine discharge zones. These authors proposed a partial analogue with similar mounds from Searles Lake in California described by Guo & Chafetz (2012), which also have prominent debris aprons produced by the weathering of tufa deposits. These reach heights now of up to 45 m, but are mainly 5 to 12 m high. The critical issue here is that these features require lake waters of similar depths to their heights, which if those in Santos Basin formed in a similar setting, requires the lakes to have been 400 m deep, although exposed at some later stage. Modern lakes in closed-basin settings can undergo remarkable fluctuations in lake level, such as the Great Salt Lake of Utah and its Pleistocene precursors (Benson *et al.*, 1992). If the Santos isolated mounds did at least in part form in a deeper lake, their textural similarities to those elsewhere in the BVF is surprising. If, as currently interpreted these textures formed close to wave base and were affected by strong evaporation (see below), their presence in a deeper setting is unusual.

Another possibility for the relief of these features seen on seismic could be that what is being seen is stratigraphic not topographic, in the sense that they appear to have relief on seismic but developed as stacked, composite low relief features, that were undergoing contemporaneous erosion around their flanks, as reflected in the associated debris deposits.

COMPOSITION

The main components of the BVF have been extensively documented from Santos Basin (Terra *et al.*, 2010; Wright & Barnett, 2015, 2020; Tanaka *et al.*, 2018; Farias *et al.*, 2019; Azerêdo *et al.*, 2021; Barnett *et al.*, 2018, in press), and equivalents of the BVF from the Campos and Kwanza basins (Saller *et al.*, 2016; Sabato Ceraldi & Green, 2017; Herlinger *et al.*, 2017; Lima & De Ros, 2019). As a consequence, what follows is only a brief summary. The typical appearance in core is of commonly heterogeneous interlayering of centimetre to metre-scale *in situ* and reworked calcite components with Mg-silicate clays (mainly stevensite and talc), with primary and replacement dolomites and silica, and porosity. What is most striking is that the formations are composed of almost only two basic components: millimetre to centimetre-sized fascicular structures (Fig. 3A and B) resembling crystal shrubs recorded from travertines (Chafetz & Guidry,

1999), producing cementstone textures associated locally with sub-millimetre to millimetre-thick isopachous laminae of coalesced fibrous calcite, and millimetre-scale radiating calcite spherulites (Fig. 3C and D).

Shrubs

Wright & Barnett (2020) record, from a study of 10 wells in the Lula area, a mean bed thickness for the shrub-bearing lithologies of 0.51 m, with 65% exhibited *in situ* shrubs and 35% as reworked shrubs. The former are identified by having their elongate crystals broadly perpendicular to the substrate, although they develop a range of radiating, branched and even plumose growth forms. The latter consists of a range of fragmented shrub material in rudstone, grainstone to wackestone and floatstone textures with or without calcite matrices. The *in situ* shrubs are separated by open pore space, or filled with Mg-silicates, or silica cement or dolomite. Where porosity is present, in some cases dolomite rhombs occur in the pore spaces like those found closely associated with the *in situ* spherulites. The shrubs grew off isopachous fibrous calcite laminae, or grew due to the asymmetrical growth of spherulites (Herlinger *et al.*, 2017; Lima & Ros, 2019), or from detrital carbonate material. The shrubs exhibit sweeping extinction (Fig. 3B), with a radial fibrous structure and also locally resemble radiaxial fibrous or fascicular optic calcite. The calcites have very high densities of very fine fluid inclusions, which impart pseudo-pleochroism. As is the case for the spherulites, some recrystallization has taken place but there is no evidence for an aragonite precursor. Such branching growth in calcite, and crystal splitting, reflect kinetic controls (Bastianini *et al.*, 2019), and are typical of precipitates from highly saturated alkaline solutions (e.g. Rainey & Jones, 2009; Jones & Peng, 2012). The range of calcite crystallographic features suggests growth from probably somewhat differing conditions and fluid compositions, but at high precipitation rates as noted for radiaxial fibrous and fascicular optic calcites (Richter *et al.*, 2011).

Wright & Barnett (2015) proposed an abiotic origin for the shrubs based on the absence of any evidence of a biological (including microbial) origin. In subsequent detailed studies this view has been validated (Herlinger *et al.*, 2017; Lima & Ros, 2019; Farias *et al.*, 2019; Hosa *et al.*, 2020). However, Sabato Ceraldi & Green

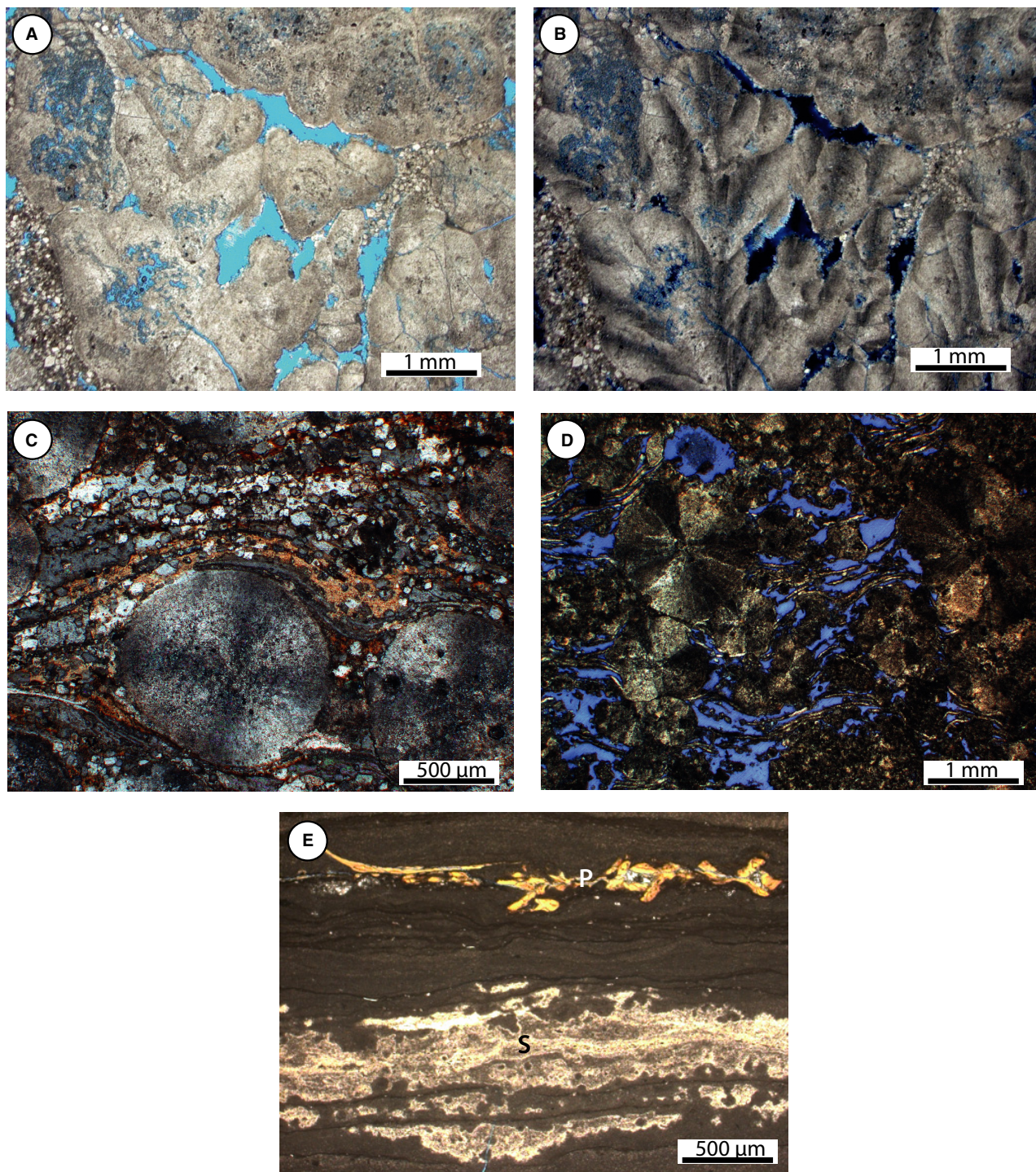


Fig. 3. Barra Velha Formation: (A) shrubs in plane polarized light showing inter-shrub porosity (blue resin) and some dissolutional porosity is present within the shrubs; (B) – as in (A) but in cross-polarized light; (C) spherulites in plane polarized light set in a Mg-silicate matrix with magnesite bridge-like structures and dolomite crystals; (D) spherulites showing magnesite bridge-like structures within the porosity (blue resin) which formed after Mg-silicate dissolution; (E) laminated carbonate mudstone (calcimudstone) showing in lower part laminae and nodules composed of microquartz (S) and compressed bone remains (P).

(2017) tentatively regarded the shrub facies from the BVF equivalents in the Kwanza and South Campos basins as microbialites. Studies of travertines with fibrous textures resembling the BVF shrubs by, for example Shiraishi *et al.* (2019) and Jones & Peng (2012) supported the view that such textures are predominantly abiotic (chemogenic) in origin. While similar textures are seen in modern active travertines, isotopic evidence suggests that the calcites were not precipitated from thermal waters (Farias *et al.*, 2019; Barnett *et al.*, 2018, in press; Pietzsch *et al.*, 2020); although data from the mounded examples described by Barnett *et al.* (2018) is not available. The lack of evidence for a microbial origin for the shrubs does not imply that the lake waters were without microbes such as extremophiles, but that there is currently no evidence that they played any role in inducing carbonate growth, although very subtle influences cannot be completely ruled out, but should not simply be assumed. The change in growth morphology from spherulites to shrubs was interpreted as indicating that the latter grew unrestricted and without a surrounding gel (see below) (Wright & Barnett, 2015; Lima & Ros, 2019), although the presence of Mg-silicates between some shrubs and also of dolomites implies that some deposition of gel may have occurred during or after shrub growth.

Spherulites

The spherulites (Fig. 3C and D) are composed of radiating fibrous calcite spheres, and reach diameters of up to 15 mm, but are more commonly <2 mm. They occur in units averaging 0.54 m thick in the Lula area wells studied by Wright & Barnett (2020), in three main textural associations. Most striking is where the spherulites occur in a clay matrix floatstone with Mg-silicate clay matrices, which possess delicate bridge-like features composed of magnesite and also numerous dolomite rhombs (Fig. 3C). These same textures occur without the clay matrices, which are now represented by porosity (Fig. 3D). These two lithofacies constitute around 60% of spherulite facies (Wright & Barnett, 2020). The remaining 40% consist of fragmented, abraded grains in rudstone, grainstone, packstone to wackestone or floatstone textures commonly associated with fragmented shrubs.

Wright & Barnett (2015) proposed that the spherulites were chemogenic and, based on published experimental studies, were the product of

rapid crystal growth in alkaline solutions, highly enriched in Mg and silica, in some form of viscous medium, which provided the support for the growing spherulites. The presence of Mg-silicate clay matrices, which formed in such fluids, originally as gels (Tosca & Wright, 2015; Wright & Tosca, 2016) supports such a model. Spherulites in the BVF in a Mg-clay matrix and/or associated with bridge-like structures, have been interpreted as *in situ* spherulites, that is, having grown within a precursor matrix (Wright & Barnett, 2020). Further studies on the Aptian spherulitic facies by Herlinger *et al.* (2017) and Lima & Ros (2019), from northern Campos Basin, and Farias *et al.* (2019) from Santos Basin, supported the explanation provided by Wright & Barnett (2015). The case for the matrices, although now Mg-clays, having originally been gels is based not only on the experimental work but also on the fact that if the matrices were formed as clays then detrital clay material, such as clay clasts, should be associated with the reworked carbonates. These are effectively never seen with some very rare exceptions. As gels, the matrices were dispersed during reworking into the water column. An analogy can be made here with flint in Cretaceous shelf chalks; these chalk muds were commonly syndepositionally reworked, but flints (originally formed as silica gels) are rarely found reworked.

However, the textural relationships and origins of the spherulites have been the subject of considerable debate (see review in Wright & Barnett, 2020). The main disagreements revolve around two issues. The first relates to the view that the spherulites did not require a gel in which to form but the case for their growth in a former gel-like matrix is addressed in detail, using textural evidence in Wright & Barnett (2020). The second issue is that many researchers have insisted that the spherulites were microbially nucleated (for example, Mercedes-Martín *et al.*, 2017; Rogerson *et al.*, 2017; Chafetz *et al.*, 2018; Kirkham & Tucker, 2018). To date no evidence has been presented for any direct microbial influence on these textures from the BVF and its equivalents, and Herlinger *et al.* (2017), Lima & Ros (2019) and Farias *et al.* (2019) have also emphasized the absence of such evidence.

The Aptian Codó Formation of the Parnaíba Basin in north-east Brazil contains microbialites and thin units with spherulites associated with Mg-clay and dolomite matrices (Bahniuk *et al.*, 2015). These authors also note the absence of

any direct evidence of microbial activity associated with the spherulites (Bahniuk *et al.*, 2015).

Microbialites

Microbialites (stromatolites, dendrolites, oncoids and microbial laminites) occur in the BVF in Santos Basin and are normally only found in the uppermost 20 to 30 m of the formation (the Lula's Fingers). An abundance of data on the frequency of microbial macrostructures and microstructures from a large set of BVF samples have been provided by Wright & Barnett (2015), with microbial macrostructures encountered in never more than 1% of thickness of logged core from the Lula area, to microscopic evidence as low as seen in 0.05% of components of thin sections. Hasiuk & Kaczmarek (2015) presented geochemical analyses from samples from Well BM-S-22 Guarani-1ST (3-ESSO-004-SPS) in Santos Basin, which they interpreted as supporting a microbial influence. This author has examined samples from that well and can confirm that microbial textures are common and readily identified, making that well highly unusual for the BVF as a whole. Muniz & Bosence (2015), from a very small data set from the Macabu Formation (the BVF equivalent) in south Campos Basin, record possible microbialites, and Hunt *et al.* (2019) also report microbialites from that basin. These studies raise the possibility that microbialites may be locally more significant in both Santos and Campos basins.

The rarity of microbial textures has been discussed by Wright & Barnett (2020) and related to the high alkalinity identified by the occurrence of the Mg-silicate stevensite, because its precipitation requires pH levels as high as 10 to 10.5 (Tutolo & Tosca, 2018). Calcification around what appear to have been cyanobacterial sheaths is seen in the microbialites in the BVF (Wright & Barnett, 2020) but is missing in association with shrubs and spherulites (e.g. Farias *et al.*, 2019). The absence of such specific microtextures in modern alkaline lakes is a consequence of high levels of DIC (dissolved inorganic carbon) (Arp *et al.*, 2001). What is also striking is the near complete absence of micropeloidal and clotted micrite textures, which are so widely documented in known and inferred microbialites. Such textures are typically interpreted as resulting from carbonate precipitation triggered by bacterial sulphate reduction, but the operation of such processes in inducing carbonate precipitation has been questioned by Meister

(2014) especially in alkaline settings (see also Gallagher *et al.*, 2013, and Meister, 2014). A preservational factor needs to be considered, as Jones & Peng (2012), documenting shrub textures (dendrite bushes) from geothermal springs in China, report that even though microbes and microbial mats were present, they had a low preservation potential in association with shrub-like growths. Those authors cite other examples of this effect from other hot spring deposits. Thus, the BVF shrubs and spherulites appear to have been chemogenic and abiotic in origin, but the lakes may well have contained microbial life, evidence of which was not preserved.

Magnesium silicates

However, the origin of the Mg-silicates needs consideration and Wright (2012) highlighted the role of microbial processes in the formation of Mg-silicates in volcanic-influenced settings. Farias *et al.* (2019) also did not exclude microbial involvement in the formation of Mg-silicates in the BVF. A number of examples of Mg-silicates associated with microbial mats have been reported (for example Jones & Peng, 2012; Burne *et al.*, 2014; Pace *et al.*, 2016; Zeyen *et al.*, 2017; Perri *et al.*, 2018; Hubert *et al.*, 2018; Jaramillo-Vogel *et al.*, 2019; Kremer *et al.*, 2019). No microbial textures have been reported associated with the Mg-silicates in the BVF (Wright & Barnett, 2020). Tutolo & Tosca (2018) have questioned whether microbial processes are capable of producing the large volumes of these Mg-silicates seen in the BVF where they can or did once constitute a high proportion of the rock volume, especially in the *in situ* spherulite facies; such occurrences can be hundreds of metres thick.

Associated carbonates

As noted above, the shrubs and spherulites are commonly not *in situ*, having been reworked. The origin of these poorly sorted and rounded grains is unclear, but suggests little winnowing or transport, and could reflect decomposition by exposure and weathering, rather than by waves or currents. While the products of this reworking are mainly poorly sorted rudstones to wackestones and floatstones, there are two distinct lithologies of particular note: laminated calcimudstones and highly texturally mature grainstones and rudstones (Wright & Barnett, 2015, 2020).

Laminated calcimudstones, composed of a fine detritus of spherulites and shrubs occur in two types. The first and more widespread (Facies 3a of Wright & Barnett, 2020) occurs in thin units, typically less than 0.3 m, of silt-grade carbonate or finer, with silt-sized quartz, clays (commonly green), lacking organic matter, finely laminated with rare desiccation cracks, locally with abundant carapaces of non-marine euryhaline ostracods, and phosphatic material including rare complete small fish (Polck *et al.*, 2020). Much of the phosphatic material consists of millimetre-long, millimetre diameter bone-like fragments (Fig. 3E). Where not compacted, these fragments are filled with carbonate matrix, and would have been hollow, these being hollow raises the possibility that they were the remains of juvenile birds or pterosaurs. Replacive-displacive pre-compaction centimetre-sized silica nodules and layers are common (Fig. 3E), composed mainly of microquartz, but with as much as 10%, very finely disseminated pyrite, commonly as fram-boids. The fine-grained nature of this facies suggests likely accumulation in a protected setting or possibly below wave base. The presence of desiccation cracks suggests periodic exposure but such features are rare. In contrast to the shrub and spherulite-bearing facies, the presence and local abundance of non-marine euryhaline ostracods suggests that the lakes may have become sufficiently less hostile to allow colonization, by fresher water inputs. The presence of well-preserved fish and other vertebrate remains is further support for this interpretation. Barnett *et al.* (2018) reported lighter oxygen isotope values for this facies (see below), further supporting the view of some influx of less evaporated, possibly meteoric waters. Wright & Barnett (2015) proposed that the early silica nodules and sheets represented the interstitial precipitation of silica gels during freshening intervals when the pH of the lake waters decreased, as is seen in some modern lakes (Renaut, 2003), although there is no evidence that this silica in the BVF was associated with microbial mats, as has been suggested for some African lakes such as Lake Magadi (Behr, 2002). The disseminated pyrite and fram-boids suggest the possibility that bacterial sulphate reduction, associated with an increase in organic activity, as indicated by ostracods and fish remains, may have occurred, which could have locally lowered the pH (e.g. Meister, 2013) to trigger silica precipitation.

The other laminated calcimudstone facies (Facies 3b of Wright & Barnett, 2020) has been

encountered only rarely to date where most of the data comes from wells that are proximal to structural highs. These decametre-thick units with varved and graded laminae, sub-metre thick graded beds resembling turbidites, lack ostracods, fish remains and early silica nodules, but are associated with thin organic laminae. This facies likely represents settings affected by sediment gravity flows.

The highly texturally mature grainstones and rudstones (Facies 5 of Wright & Barnett, 2020) have been documented by Barnett *et al.* (2018, in press) and occur in decametre-scale units composed of well-sorted and well-rounded grains of shrubs, spherulites, intraclasts and, in some cases, peloids and volcanic fragments. These grainstones and rudstones exhibit sharp-based, decimetre to metre-scale fining-upward packages with, planar/low-angle stratification, and less common cross-lamination and planar cross-bedding. This facies type is associated with marked thinning of the BVF around margins of tilted fault blocks and is interpreted as wave-dominated fan delta shoreline deposits by Barnett *et al.* (2018, in press), and possibly spits.

Aspects of the limited range of components

There are several outstanding features about the composition of the BVF that require some attempt to explain. In most wells there is a paucity or even lack of what one might regard as normal carbonate components such as micrite and sparite. In addition, considering the abundance of calcite precipitates, ooids are extremely rare. A likely explanation is that carbonate nucleation was suppressed by kinetic inhibition favouring growth at existing sites (Wright & Tosca, 2016; see also Bastianini *et al.*, 2019).

The main primary components of the BVF were calcite and Mg-silicates (especially stevensite, a trioctahedral smectite) but if evaporation played a critical role in producing some or all of the key components (see below), the basic question needs to be asked as to why more typical evaporite minerals are missing, especially sulphates and chlorides? This has been addressed by Wright (2012) citing Cerling (1994) who, using data from lakes in the East African Rift, emphasized that volcanic catchments are poor in sulphates and chlorides.

Cerling (1994) proposed that where volcanic terrains predominate, calcite, tri-octahedral smectite, analcime and various bicarbonate-carbonate minerals would be the main primary

minerals, and the outboard rifted settings of the South Atlantic Cretaceous basins would provide such a setting. The absence of sulphates such as gypsum could be explained if the $[\text{HCO}_3^-]$ of the lake waters was very high relative to $[\text{Ca}^{2+}]$ as may have been the case in the Archean (Grotzinger & Kasting, 1993), and the influence of high pCO_2 levels will be discussed later. Mercedes-Martín *et al.* (2019) have suggested that the absence of sulphates and chlorides in the BVF lakes reflects the effects of stationary leakage flux to the underlying aquifers, while the model proposed by Farias *et al.* (2019, fig. 18) of seawater seepage would have produced sulphates and chlorides in the BVF lakes.

Finally, siliciclastic material is uncommon (see also Gomes *et al.*, 2020), suggesting that the likelihood of any sediment from the adjacent continents was trapped before reaching the lakes. Volcanic clasts are present in the well-sorted grainstone facies (Barnett *et al.*, 2018).

Diagenesis

The diagenesis of the BVF-type facies has been discussed in general terms by Tosca & Wright (2015), who suggested that much of the diagenesis could relate to the decay of stevensite. In Santos Basin, a prominent phase of later stage carbonate dissolution took place enhancing porosity and permeability in the reservoirs, after several phases of replacement and cementation by silica and dolomite. A contributor to this dissolution might have been the decay of the Mg-silicates, which can result in the release of hydrogen ions (Tosca & Wright, 2015).

The role of the salt cover promoting geothermal convection, during and after salt deposition has been investigated by Jones & Xiao (2013) using reactive transport modelling, raising the possibility that some of the dissolution could have been driven by such convection. Detailed studies by Herlinger *et al.* (2017), Vieira de Luc *et al.* (2017), Tritlla *et al.* (2018), Lima & De Ros (2019) and Lima *et al.* (2020), have emphasized the extensive hydrothermal overprint seen in the Campos Basin relative to that in the Santos Basin. Saller *et al.* (2016) and Teboul *et al.* (2017) have also assessed the role of such effects in the Kwanza Basin.

Classifying the Barra Velha rock types

Providing a robust classification for these carbonates presents several problems. As discussed

by Wright & Barnett (2020), the growth of the spherulites within a Mg-silicate matrix (see also Herlinger *et al.*, 2017; Lima & De Ros, 2019) creates an unusual type of texture, which after the congruent dissolution of the Mg-silicate of the matrix (Tosca & Wright, 2015) leaves what apparently could be described as a porous grainstone or even boundstone. To address some of these terminological problems Wright & Barnett (2020) have proposed a simple classification that includes consideration of the porosity, while Gomes *et al.* (2020) have offered a more detailed system and also discuss non-carbonate detrital components. Azerêdo *et al.* (2021) also provide another approach to classifying the textures in the BVF.

CYCLICITY AND DEPOSITIONAL SETTING

As stated earlier, the distinctive suite of lithofacies characterizing the BVF and its equivalents have been identified in Santos, Campos and Kwanza basins. Besides the lack of any analogue for this scale of chemical sedimentation, the lack of sedimentary indicators of water depth has hampered initial attempts to define the depositional setting. A critical step in interpreting the BVF came with the discovery of cyclothems in some wells (Wright & Barnett, 2015) (Figs 4 and 5). These authors noted a statistically tested pattern of repetition of three lithologies in metre-scale units: ostracod-bearing laminites (facies 3 of Wright & Barnett, 2015), overlain by *in situ* spherulite (facies 2), overlain by *in situ* shrubs (facies 1).

Wright & Barnett (2015) proposed that the cyclothems likely represented flooding and evaporation cycles. Initially an influx of water, probably as run-off, deepened the lakes to below wave base allowing fine-grained carbonate to accumulate. It is important to note that wave base in lakes is commonly shallower than in marine systems because of the reduced fetch. This influx also lowered the alkalinity–salinity sufficiently for ostracods and vertebrates to colonize the lakes, these were likely to have been introduced from more permanent water bodies such as nearby streams or other lakes. The reduced alkalinity triggered silica precipitation (see above). The overlying spherulite-bearing units were first deposited as Mg-silicate gels (Tosca & Wright, 2015; Wright & Tosca, 2016), requiring a shift to highly alkaline waters with

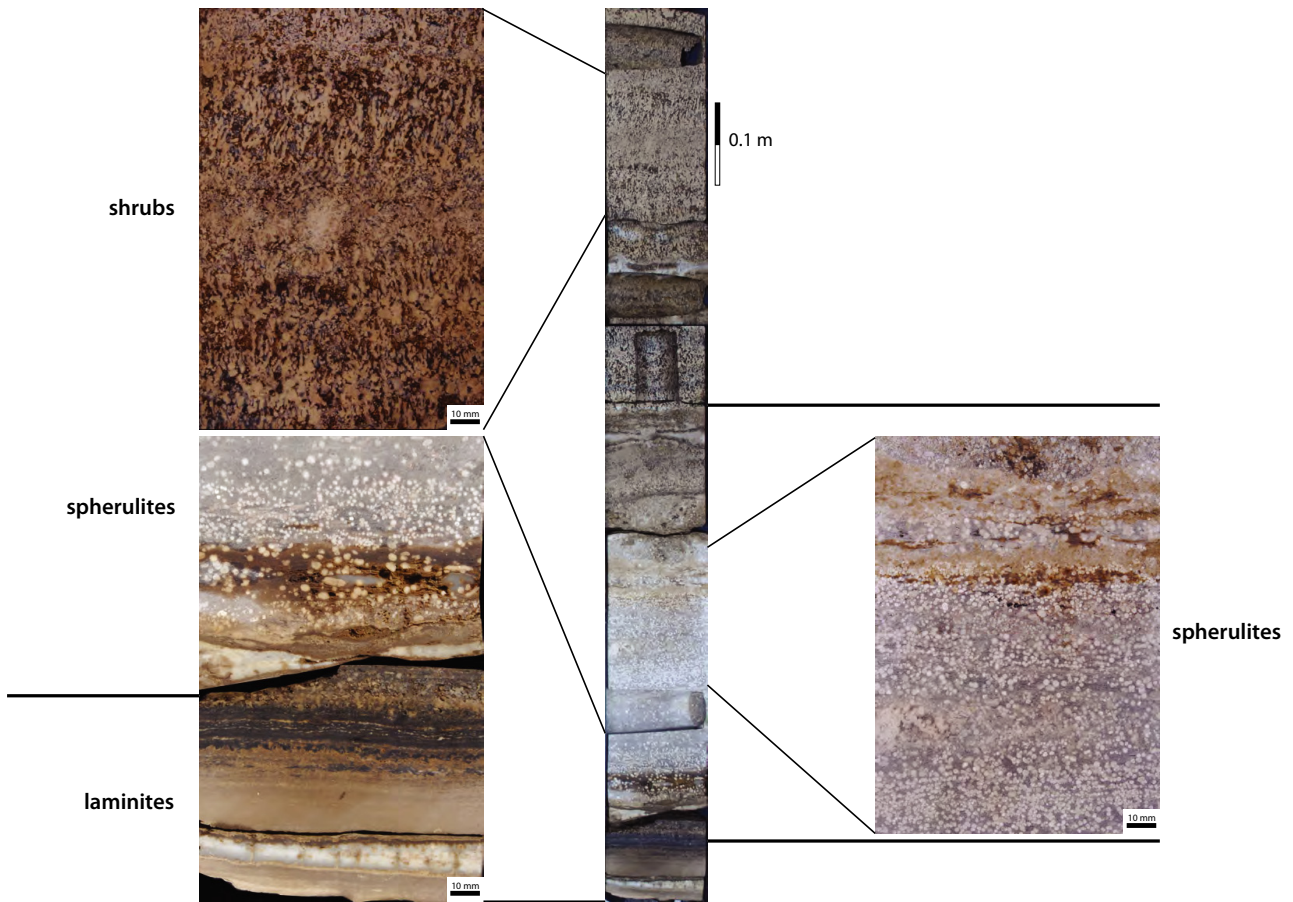


Fig. 4. Core image of cyclothem in Barra Velha Formation; top of cyclothem is 0.4 m above top of core stick.

Cyclothem 0.75–5 m thick

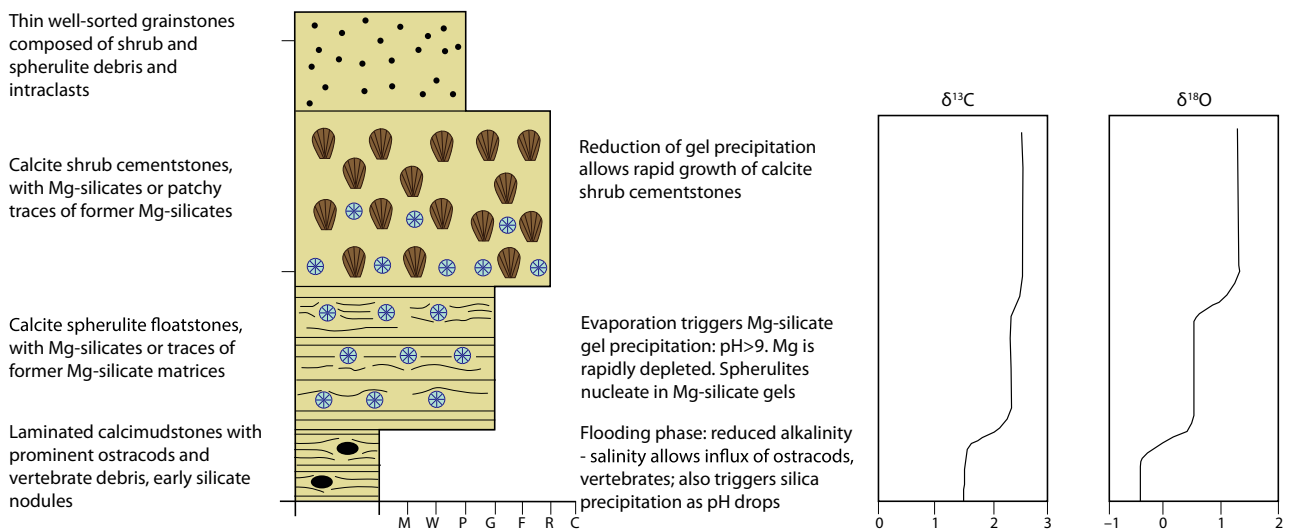


Fig. 5. Typical cyclothem found in the Barra Velha Formation from Santos Basin (based on Wright & Barnett, 2015; Barnett *et al.*, 2018, in press).

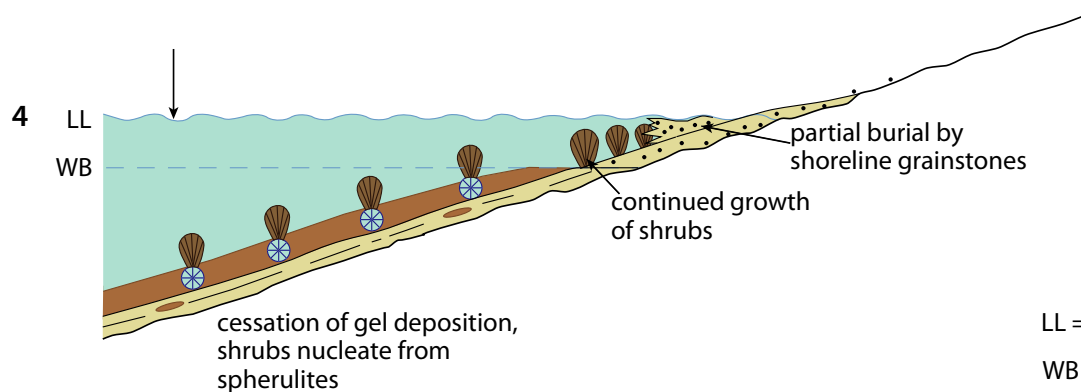
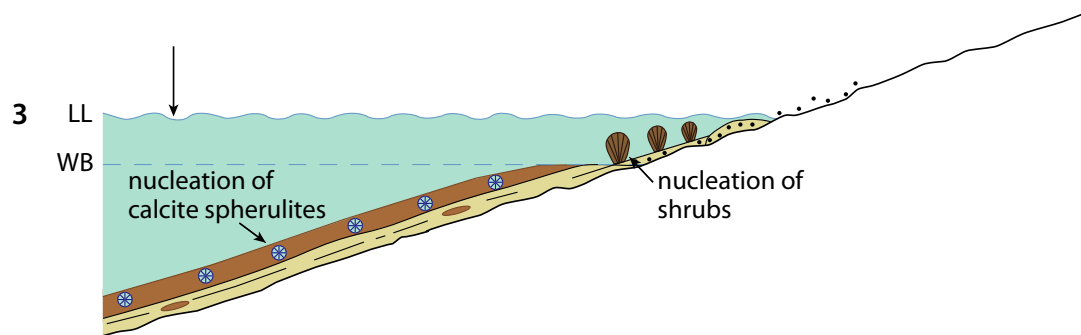
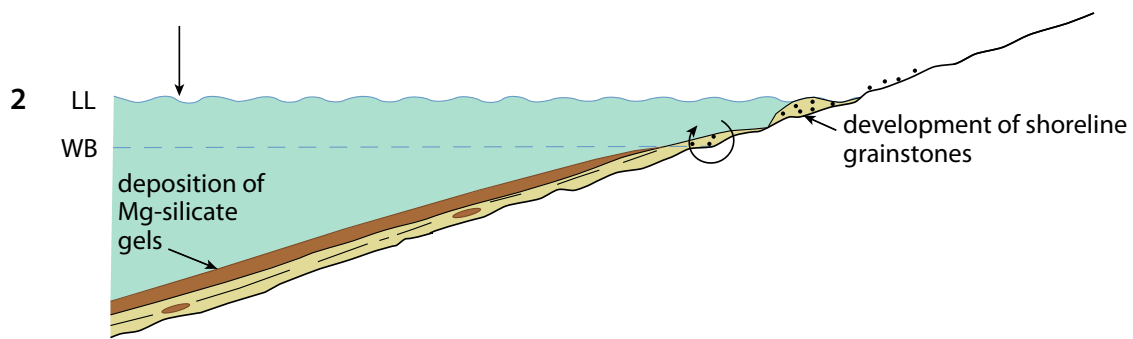
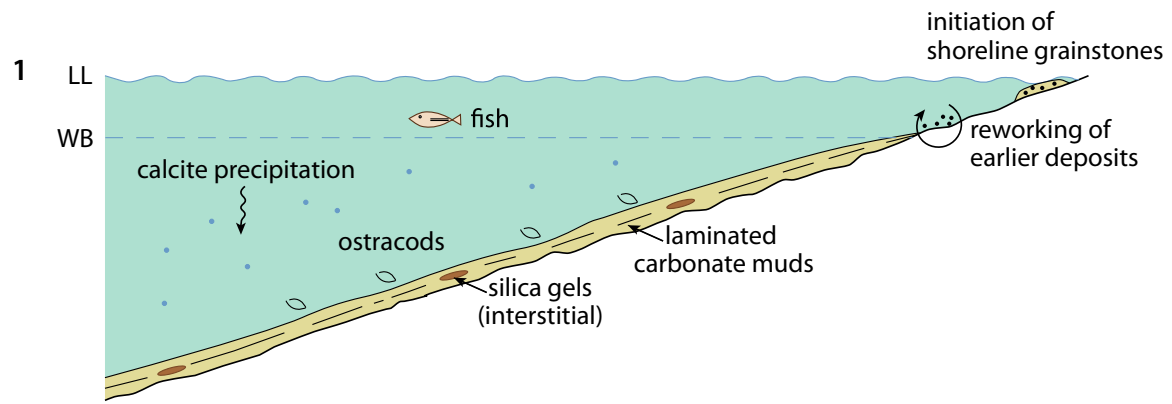
pH of 10 or above. The spherulites, Wright & Barnett (2015) proposed, grew in this gel and in many cases seeded the shrubs in the overlying shrub layer (see also Herlinger *et al.*, 2017; Lima & De Ros, 2019; Farias *et al.*, 2019). Most gels would likely have been deposited and accumulated below wave base but were also likely to have been periodically reworked. At some stage, gel precipitation stopped or decreased, possibly because of Mg depletion, or changed pH levels (Wright & Barnett, 2015). Many of the *in situ* shrubs have open porosity but, as noted above, there is evidence that some shrubs likely had Mg-clay matrices (Wright & Barnett, 2015). An explanation could be that during the growth of the shrubs any gel material would have been removed by wave action as a result of progressive shallowing. Finally, in the upper part of the cyclothems, shrub growth also ceased. This could have been triggered by several mechanisms, such as decreases in CO₂ content of groundwaters (e.g. Jones & Peng, 2012). In some cyclothems, the shrub units are overlain, seemingly non-erosively, by centimetre-thick, well-sorted grainstones composed of shrub and spherulite debris, then sharply overlain by laminites.

Support for this evaporation model came from thermodynamic modelling (Wright & Tosca, 2016), which indicated that the progression from Mg silicate formation to calcite precipitation could result from concentration by evaporation. Barnett *et al.* (2018, in press) have provided C and O stable isotope profiles for the cyclothems (Fig. 5), which indicate a progressive enrichment in C¹³ and O¹⁸ during the cyclothem, commensurate with an evaporation trend. If, as this data suggests, the carbonate laminites have lighter values than those from the spherulites or shrubs, which

contain fine fragments of these grain types, some lighter carbon was added, which might imply that carbonate was precipitated during that freshening phase. Pietzsch *et al.* (2020) from a single well data set from Santos Basin, note a: “lack of direct correspondence between lithofacies and isotopic composition” but suggest that evaporative conditions prevailed including the suggestion that inflow waters became enriched in δ¹³C due to evaporation.

One problem with this evaporative model relates to the known distribution of shrub-dominated facies. If the shrub facies represents the more evaporative phase of the lake, it should, based on existing models for the distribution of lake evaporites (for example, Renaut & Gierlowski-Kordesch, 2010), be limited to those parts of the system where more evaporated waters would have pooled, that is, in areas with lower topographies (hence the bulls-eye model for lake evaporite distribution). However, shrub facies are often best developed in more up-dip areas [for example Barnett *et al.* (in press), fig. 12]. Figure 6 is an attempt to explain this apparent anomaly by considering also the effects of changing water depth and wave action as lake waters fell. However, this model might also produce shrubs with different C and O isotopic values having begun forming during different stages of evaporation, or suggest that individual shrubs might also show such changes. If the shrub facies represented the peak of evaporation it is noteworthy that such units, from this author's experience, do not display evidence of erosion or exposure, but Gomes *et al.* (2020) note that a key feature of the tops of these cyclothems are irregular surfaces which: “may or may not have been subaerially exposed”.

Fig. 6. A model for the development of cyclothems in the Bara Velha Formation. *Stage 1:* filling of the lake by waters of reduced alkalinity–salinity, introducing conditions conducive to euryhaline ostracods and fish. The lowering of the pH triggered precipitation of interstitial silica gels which later formed silica nodules. The lighter oxygen isotope values in this unit suggest precipitation of some calcite from these less evaporated waters as compared with the values found in the spherulites and shrubs. The accumulation of fine-grained carbonates implies deposition in a lower energy setting, probably below wave base. Earlier deposits were reworked in shallower areas along wave-dominated shorelines. *Stage 2:* represents the point when evaporation triggered the formation and deposition of Mg-silicate gels, which are assumed to have been deposited below wave base. During the contraction of the lake it is assumed that wave base may also have been shallower as fetch decreased. *Stage 3:* the precipitation of calcite, as spherulites within the gels below wave base, and locally updip as shrubs where gels were not accumulating or had been removed by falling wave base. This is based on the observations that shrub facies are better developed, with commonly larger shrubs, in updip areas near minor palaeo-topographic highs. *Stage 4:* gel deposition may have ceased and spherulite growth reached the sediment–water interface allowing shrubs to grow; further shrub growth took place in shallower settings. Later local progradation of shoreline grainstones covered the shrubs capping the cyclothems.



LL = lake level

WB = wave base

Azerêdo *et al.* (2021) describe the effects of early dissolution on the shrub facies from the BVF, possibly linked to their disintegration, which might support the view that this shrub phase terminates each evaporation cycle. There have been other explanations for the 'spherulite to shrub' change besides evaporation. For example, Tosca *et al.* (2018) suggested that the 'spherulite to shrub' transition could have arisen due to water column stratification, triggered perhaps through freshwater influx to a dense, saline water body because they considered that the spherulites required higher states of saturation than shrub growth based on the effects of diffusive fluxes and nucleation rates on crystal growth. The role of stratification and its breakdown by mixing and upwelling has been invoked for large-scale chemogenic precipitation in Archean to Palaeoproterozoic ocean basins by Grotzinger & Knoll (1995).

The distribution of the shrubs might be better explained if they formed during periods of rising lake level and Gomes *et al.* (2020) have suggested that wetter periods in the basins, with rising or higher lake, would have reduced alkalinity and favoured shrub formation. This does not explain the isotopic data shown by Barnett *et al.* (2018, in press). Gomes *et al.* (2020) raise another important issue in proposing that, besides the commonly identified trend for an upward increase in shrubs (Wright & Barnett, 2015; Farias *et al.*, 2019; Lima & De Ros, 2019), they also identified an upward-increasing spherulite trend. These authors have also incorporated changes in the detrital content and have offered several possible scenarios to explain facies changes: humid to arid trend invoking the evaporative model (upward increase in shrubs); semi-arid–arid model whereby the shrubs formed during a less evaporitic phase, followed by further evaporation by spherulites (upward-increasing spherulite trend), and lastly a constant lake level model whereby changes such as those seen in the main components were the result of changes in aeolian/fluvial detrital input, pH and lake stratification, induced by climate change.

The second model as noted by Gomes *et al.* (2020) is related to that proposed by Farias *et al.* (2019) from a study of a small (50 m) data set from one Santos Basin well (SB-2). These authors suggested that the laminite facies formed during a flooding stage, followed by evaporation resulting in shrub growth, and Mg-silicate precipitation. Further evaporation led to complete desiccation (Farias *et al.*, 2019, fig. 14)

and the growth of displacive spherulites, and apparently halite precipitation. However, this mechanism does not explain the widely reported fact that many shrubs nucleated off spherulites, or the lack of desiccation features in the spherulite facies, and that the calcite pseudomorphs after halite reported by these authors appear to be dolomite (compare fig. 13 of Farias *et al.*, 2019 with those of similar textures described as dolomite in Saller *et al.*, 2016; Herlinger *et al.*, 2017; Lima & De Ros, 2019; Wright & Barnett, 2020; Gomes *et al.*, 2020).

Gomes *et al.* (2020) reinforce the point inherent in the application of an evaporation-driven model (Wright & Barnett, 2015), that the spherulites (with Mg-silicates) formed during particular stages of lake development. This raises a conceptual problem if one interprets them as contemporaneous, as is implied in the models presented by Saller *et al.* (2016), de Paula Faria *et al.* (2017) and Sabato Ceraldi & Green (2017).

Tanaka *et al.* (2018) also offer a facies model that identifies the shrub facies as stromatolites, which formed in shallower settings than spherulites. However, these authors relate the laminite facies to the shallowest settings and make no reference to its biota. This model is based on the brief interpretations provided in de Paula *et al.* (2017).

Modern closed system lakes such as that envisaged for the BVF facies have three potential water sources: run-off, direct precipitation and groundwater recharge. The importance of these inputs varies between different lakes and also with time, related to climate changes. The deposition of the laminites (Facies 3 of Wright & Barnett, 2015) suggests that some local run-off took place, bringing in little non-carbonate sediment, but able to dilute the lake waters and introduce vertebrates and ostracods. As regards evidence for large-scale groundwater injections, to date no clear examples of vents have been described from the BVF, although in Campos Basin some seismic features, including what appear to be isolated mound-like build-ups in the earlier rift stage, have been interpreted as hydrothermal vent complexes by Alvarenga *et al.* (2016), and the isolated high relief mounds in Santos Basin described by Buckley *et al.* (2015) and Barnett *et al.* (2018), discussed earlier, might be related to sub-lacustrine vents. However, in modern lakes groundwater input can be significant, such as in some modern rift lakes such as Nasikie Engida (see above), and Farias *et al.* (2019) have used clumped isotope

analyses to suggest that the BVF carbonates recorded in Well SB-2 formed in the range 46° to 73°C, so are likely the products of thermal springs. Pietzsch *et al.* (2020) concluded, using standard C and O isotopic data from the BVF that the isotopic signal for the BVF carbonates was from evaporation of lake waters that had undergone long residence times. If the carbonate and silica-charged fluids had been sourced via thermal vents, those waters were likely not precipitating near the vents.

Mercedes-Martín *et al.* (2019) have argued that the evaporation model proposed by Wright & Tosca (2016), based on the cyclothems described by Wright & Barnett (2015), could not be produced simply by evaporation from a single freshening–evaporite cycle. Those authors used data from Lake Baringo to show that 25 to 50 fluctuations would be required to produce the cycles. While suggesting that such metre-scale cyclothems reflected evaporation (as supported by the isotope data in Barnett *et al.*, 2018, in press), Wright & Tosca (2016) did not imply that the cycles were the product of a single event. Here an analogy can be made, with suggesting that a peritidal carbonate cyclothem represents the product of one tide. However, Wright & Tosca (2016) did not provide an explanation as to the hydrological setting that could generate such metre-thick cyclothems, which apparently indicate a simple, if prolonged, evaporation trend. Mercedes-Martín *et al.* (2019) suggested that the facies cyclicity recognized in BVF largely reflects temporal fluctuations in the type of waters sourcing the lake and in the degree of leakage to aquifers, rather than variations in the lake water concentration factor due to evaporation. If groundwater were a major source of the lake water, thermal or not, what could be being reflected in the cyclothems is initially a freshening (deepening) phase of the BVF lakes, likely a result of input from run-off (introducing fish and ostracods from streams or other refugia), followed by the increasing effect of water derived from groundwater input, plus evaporation. This could be readily tested by analysing the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of each carbonate phase to detect changes in the provenance.

Intriguing as these important cyclothems are, they are not the dominant features of the BVF, which consists of a highly heterogeneous mix of *in situ* and reworked components, often in poorly rounded and sorted grain–wackestone textures, with ill-defined units lacking evidence of exposure and diagnostic sedimentary

structures, except for the mature grainstones, interpreted as fan deltas and spit deposits as documented by Barnett *et al.* (2018, in press). If these *in situ* and reworked lithofacies are reflecting formation under different water depths, with or without changes driven by evaporation, they are indicating a highly dynamic system. This is not unexpected in shallow, closed lake systems where water depth changes, over geologically short time intervals, can range from decametres or more, over hundreds of years, to metres on a decadal scale or even less. These changes reflect fluctuations in rainfall, snow melt (although not in the case of the BVF), and hence run-off or groundwater discharge, evaporation rates and of changes in the natural connections between nearby lake segments. Metre-scale fluctuations are commonly encountered around lakes due to variations in wind direction. The Pleistocene to present day history of the Great Salt Lake in Utah being such an example of the remarkable changes that can occur in water depths, at different scales (Gwynn, 1998). The variability in styles of lithofacies ordering in the BVF, as shown by Wright & Barnett (2015), Gomes *et al.* (2020) and Azerêdo *et al.* (2021), and the general heterogeneity, is likely an indicator of such restricted conditions and of the interplay of minor topographic variations with complex hydrological and physical factors, including stratification.

It is clear that the origins of the BVF facies are not the subject of universal agreement, whether in relation to the sources of their components, or to the gross depositional environment, as well as the hydrological implications of the cyclothems. In broadest terms two somewhat contradictory models can be invoked (Fig. 7): the deep lake model, based mainly on seismic facies (see above); and the shallow lake model based on facies analysis, geochemical evidence (C and O isotopes) and correlation studies. The task of deciding which, if any, model works, and indeed whether that model would be expected to work over such a huge area, is hampered by the lack of any modern (or ancient) analogue (including seismic analogues for possible large lacustrine build-ups), the paucity of depth indicators and the biased nature of the data coming from structural highs, let alone the structural complexity of these basins. However, there has been an imperative for such models coming from the exploration for oil in these basins.

Whatever large-scale model is invoked, the current consensus is that the carbonate factory

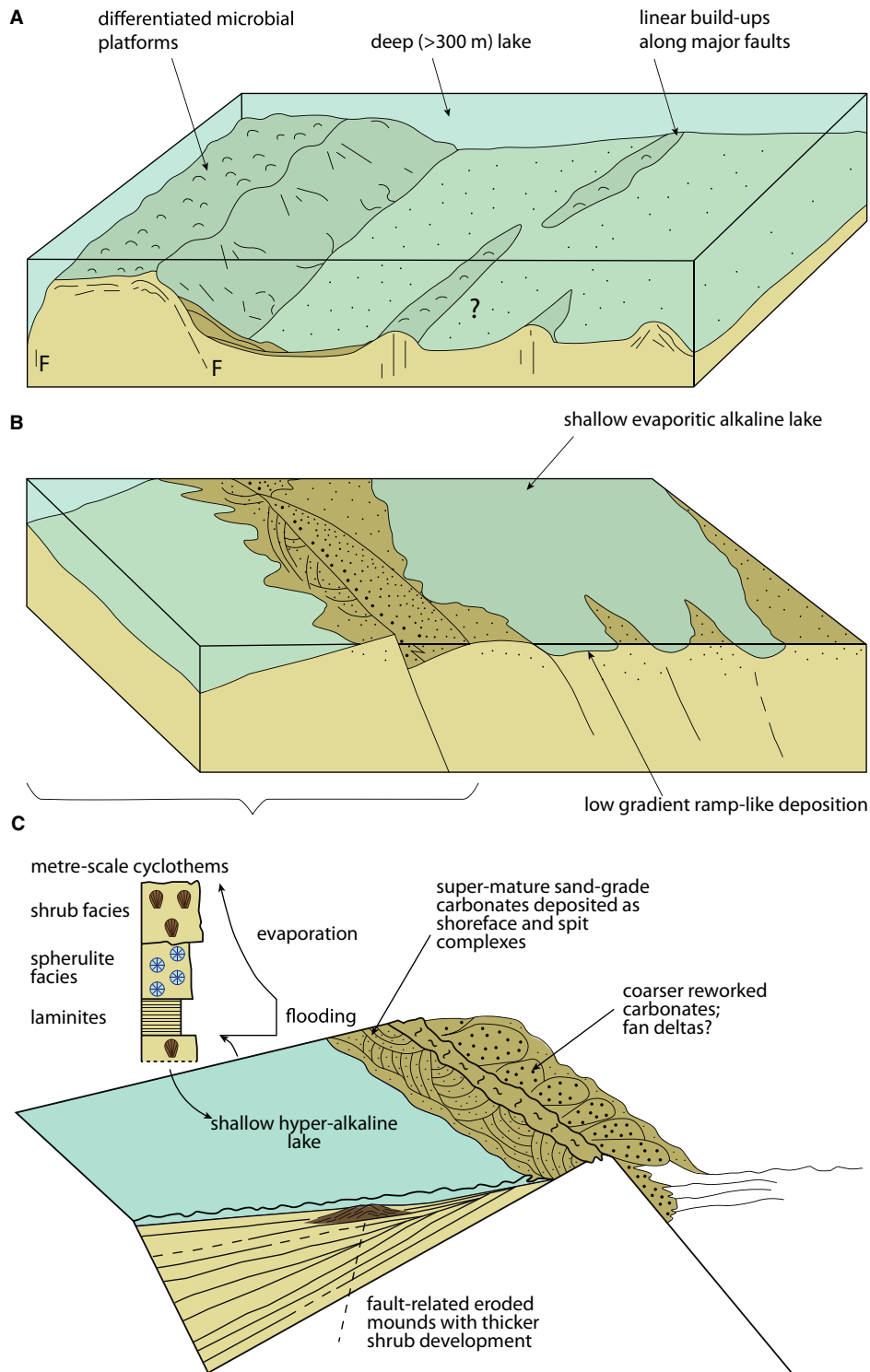


Fig. 7. The use of the scale-less obligatory block diagram to help visualize the broad depositional models invoked for the Barra Velha Formation and its equivalents. (A) *The deep lake model*, differentiated platform model invoked by various authors (see text), in which estimates of water depths have been based on interpretations of dipping reflectors as carbonate slope clinoforms. (B) *The shallow lake model*, with possible facies distributions shown in (C) at greater detail.

was overwhelmingly abiotic, that is, chemo-genic. If extremophiles were present, as seems likely even in such alkaline conditions, the evidence for their involvement, direct or indirect, in carbonate precipitation is lacking and, no matter how topical it is to invoke many carbonate features as the product of microbial processes or the influence of microbial substrates such as extracellular polymeric substances (EPS), any evidence-based interpretation must currently favour an abiotic origin. This makes the vast BVF and its equivalents a somewhat rare, almost unique, carbonate factory in Earth history, and certainly in the Phanerozoic.

DISCUSSION: THE BVF FACTORY

The early (Aptian–Albian) Cretaceous is regarded as an interval when high rates of volcanic CO₂ in the atmosphere led, via high rates of silicate weathering, to high rates of shallow water carbonate production in the oceans (Pohl *et al.*, 2020). However, the lakes which produced the BVF and its equivalents were not linked to the Aptian oceans. If such enhanced weathering from elevated levels of atmospheric CO₂ had supplied the BVF factory, the limited range of precipitates suggests a predominantly basic igneous source area, yet little detrital material from such a source, or any major non-carbonate source, entered the deposystem; therefore subaerial weathering seems an unlikely process generating the chemistry of the lakes.

If a consensus exists in the studies on this chemistry, it is that these lakes were at times hyper-alkaline, with pH of 10 or even above. This alkalinity requires explanation, an issue in other ancient hyper-alkaline lake systems as noted by Lowenstein *et al.* (2017), who identified several possible mechanisms whereby hyper-alkalinity could arise. Periods of high atmospheric CO₂ levels triggering accelerated weathering could provide the alkalinity, as noted by Farias *et al.* (2019), potentially enhanced by the sub-sea-level elevation of the lake basins. Magmatic activity introducing alkaline Na⁺–HCO₃[–]-rich hydrothermal waters is another mechanism and operates within basins such as Lake Magadi, Kenya, or at distance such as Searles Lake, California (Lowenstein *et al.*, 2016). Lowenstein *et al.* (2017) also noted that high levels of organic decay in lakes could provide a source for CO₂-related alkalinity, but adding that such organic activity in modern lakes

had not created hyper-alkalinity. In fact high input of CO₂ producing high levels of dissolved inorganic carbon has been invoked more recently as potentially promoting high organic productivity, carbonate formation, alkalinity and the production of Mg-smectites in some lakes (Milesi *et al.*, 2019, 2020).

Milesi *et al.* (2019, 2020) and Wright & Barnett (2020) suggested that inputs of magmatic CO₂ into the lakes represented by the BVF carbonates could have influenced carbonate and non-carbonate deposition by creating extreme carbonate and bicarbonate alkalinities. High alkalinities are common in evaporitic lakes where magmatic CO₂ input is high (Pecoraino *et al.*, 2015). Carbon dioxide is present in these South Atlantic basins of Brazil and, in one field, CO₂ was found to be up to 80% in the gas cap (Gamboa *et al.*, 2019). The bulk of the CO₂ in these Atlantic margin basins of Brazil has been shown to be of mantle origin (e.g. Santos *et al.*, 2012), and that wells located near major intracrustal intrusions contain the highest amount of CO₂, which has been interpreted as likely sourced from the mantle during late Cretaceous and early Eocene times (Gamboa *et al.*, 2019). The volcanic activity that was associated with the deposition of the BVF and its equivalents (Szatmari & Milani, 2016) would have likely provided an additional source for CO₂ and carbonate alkalinity. Lima *et al.* (2020) have discussed the possibility that CO₂ could have been supplied from the upper mantle or serpentinization.

Szatmari & Milani (2016), while suggesting that leaching by phreatic and hydrothermal waters and erosion of flood basalts supplied Ca²⁺, Mg²⁺ and SiO₂ for the BVF facies, identified mantle CO₂ released later in the Cretaceous and Eocene as the cause of hypogenic leaching to produce reservoir quality. However, CO₂ from the mantle was likely actively entering the BVF lakes, and also introducing leachate as bicarbonate from underlying units, during formation of the BVF. Pietzsch *et al.* (2020) discuss in detail the isotopic fractionation, composition, trends and origins of C and CO₂ in the BVF, including a mantle origin. These authors note that in the BVF lakes equilibrium took place between lake water DIC reservoir and atmospheric CO₂, that required long water residence times coincident with evaporitic shallow lake waters. Studies of the active CO₂-rich Lake Dziani Dzaha, Mayotte, Indian Ocean (Milesi *et al.*, 2019, 2020) have shown that magmatic CO₂ provides a large contribution to the dissolved organic carbon in the

lake, promoting primary productivity, high alkalinities, carbonate formation and the formation of Mg-silicates, albeit on a smaller scale than the BVF lakes.

There is a case that the alkalinity in the BVF lakes was at least influenced by magmatic and mantle CO₂, with that CO₂ both contributing directly to the lake waters, and also as an agent for leaching a mafic-rich source to provide the elemental suite required for the mineralogy of the BVF. As stated above, the BVF lakes lacked non-carbonate terrigenous input other than minor amounts of silicates, which suggests that subaerial weathering was not the source of the lake chemistry. Here ⁸⁷Sr/⁸⁶Sr data provides insights into the provenance of that chemistry. Pietzsch *et al.* (2018), Farias *et al.* (2019) and Obermaier *et al.* (2019) provide ⁸⁷Sr/⁸⁶Sr data from carbonates that shows values significantly higher (more radiogenic) than Cretaceous marine values. The more radiogenic values in the BVF preclude either the involvement of a mantle or oceanic basalt source, and that serpentinization was likely not involved in sourcing the BVF carbonates. Because the ⁸⁷Sr/⁸⁶Sr ratios are too radiogenic compared with continental flood basalts in the region, Pietzsch *et al.* (2018) suggest that felsic rocks from the basement and/or alluvial material derived from it were likely also contributing ions. As Pietzsch *et al.* (2018) acknowledge, the lack of significant detrital sediment in the BVF suggests that direct input from erosion and surface transport from local catchments and rivers seems unlikely, whereas the radiogenic values could reflect waters that had long residence times in acidic bedrock aquifers, as noted by Ojiambo *et al.* (2003) from east Africa. These authors present a model using mass-balance calculations to show that a felsic source had likely contributed with fluxes one to two, and up to four, orders of magnitude higher than the mafic source. This would likely require sourcing from the Precambrian basement. Tutolo & Tosca (2018) have suggested that waters with high concentrations of silica and alkali cations typical of intermediate to felsic rocks, such as Na and K, would be needed to produce the Barra Velha mineral suite. However, Lima *et al.* (2020) raise the problem that invoking a felsic source, such as the underlying granitic-gneissic basement, would not provide a suitable source for the abundant Mg and Ca in the BVF and its equivalents.

Pietzsch *et al.* (2018) also identify a trend towards more radiogenic values from the Jiquiá

to Alagoas stages (Itapema to BVF), while Obermaier *et al.* (2019) report a clear shift to more radiogenic values across the pre-Alagoas unconformity, and that the overlying evaporites of the Ariri Formation indicate a trend to more marine values. Lima *et al.* (2020) provide a compilation of ⁸⁷Sr/⁸⁶Sr data from Campos and Santos basins that also shows the trend towards more radiogenic values into the Barra Velha and Macabu formations. This more radiogenic signature in the BVF relative to the Itapema contrasts with the interpretation by Farias *et al.* (2019) that the BVF represents an: “initial/transitional stage ending with the deposition of halite and bittern salts” of the marine-related Ariri Formation.

Lima *et al.* (2020) report that, as isotopic data from different studies were similar over a regional scale in Campos and Santos basins, this raises the possibility that the BVF represents either single vast lake or lakes that were hydrogeologically connected.

The possibility of some involvement of serpentinization of exhumed mantle requires some discussion, because some travertine carbonates are associated with exposed ophiolite complexes and serpentinization (e.g. Leleu *et al.*, 2016). Pinto *et al.* (2017) have asked whether the Ca and Mg necessary to build the Aptian carbonates in the region were related to fluids derived from serpentinization. Zalán *et al.* (2011) identified an extensive zone of exhumed mantle in Santos, Campos and Espírito Santo basins, and proposed that serpentinization had taken place, but others have questioned the exhumation of the mantle in the region (Kumar *et al.*, 2012; Evain *et al.*, 2015). Pindell *et al.* (2018, fig. 6) implies that exhumation occurred syn-salt deposition and not before. The ⁸⁷Sr/⁸⁶Sr ratios of the BVF carbonates do not indicate a mantle or oceanic basalt source, but Lima *et al.* (2020) have invoked the possibility of some involvement of fluids related to serpentinization to explain the lower ⁸⁷Sr/⁸⁶Sr ratios observed in the later hydrothermal diagenetic phases compared with those of the primary carbonates in the BVF facies in Campos Basin.

On an even larger scale, the relationship of the BVF lake chemistry to changes in global CO₂ and OAE 1a (Oceanic Anoxic Event 1a) has also been discussed (Tedeschi *et al.*, 2017; Pietzsch *et al.*, 2020), and the issue arises as to whether the BVF lakes were passive witnesses to that event or that magmatic (mantle) CO₂ and its release from the formation of the BVF carbonates

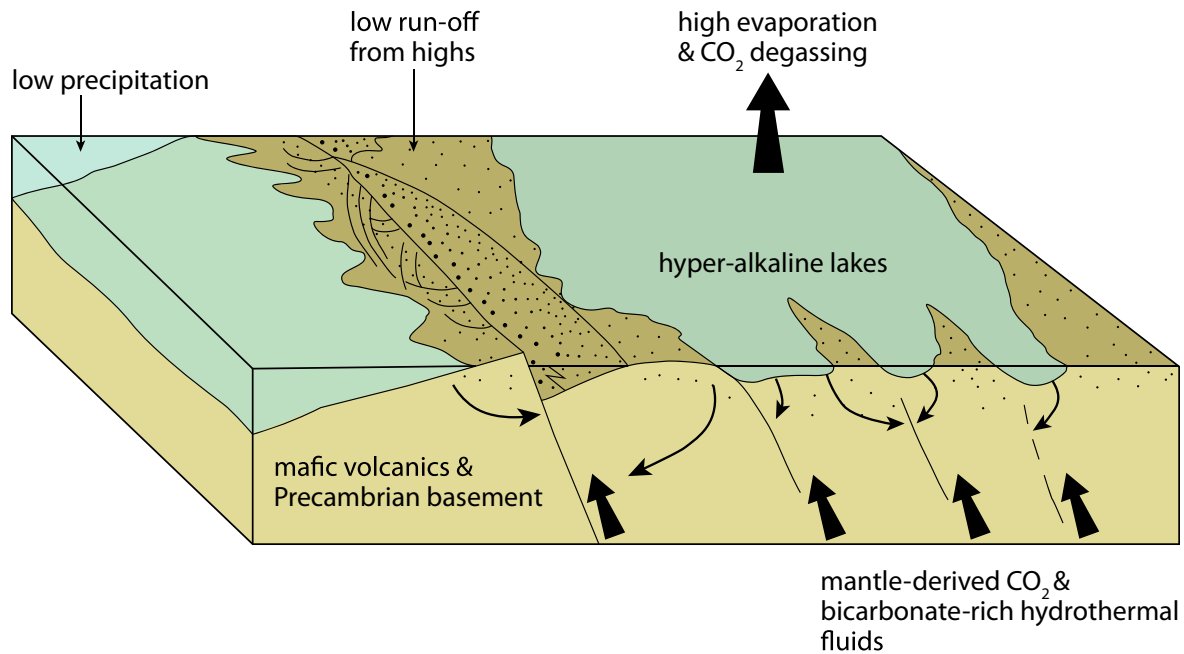


Fig. 8. An attempt to illustrate the hydrology of the Barra Velha Formation, emphasizing the role of mantle CO_2 and fluids in producing this unique but areally-extensive non-marine carbonate factory.

had any impact on that event? Tedeschi *et al.* (2017) have provided arguments in favour of deposition of the topmost part of BVF at the onset of OAE 1a. Pietzsch *et al.* (2020) suggest that the negative $\delta^{13}\text{C}$ trend observed in the upper part of Well S10 in Santos Basin would correspond to the onset of the OAE 1a, and increased atmospheric CO_2 , and might have favoured the predominance of the microbial sedimentation instead of inorganic precipitation in the upper BVF 3 (as is seen in the Lula's Fingers, see above). There is uncertainty over the exact ages of the BVF (see *Geological setting*) and of OAE 1a (Jenkyns, 2018). The increase in atmospheric CO_2 associated with OAE 1a has been attributed mainly to sustained mantle eruptions by Adloff *et al.* (2020), and has been directly attributed to volcanism related to the Ontong Java Plateau (e.g. Erba *et al.*, 2015). It remains to be determined if the vast scale of the BVF might imply an additional, South Atlantic source for this CO_2 .

Some fundamental issues relating to the basic hydrology of the lakes remain unclear. Where did the water come from, because there is little evidence of run-off transport other than the carbonate dominated fan deltas inferred by Barnett

et al. (in press)? Could water have been provided by direct precipitation or were vents a more likely source? To date no examples of travertine or tufa facies have been described from the BVF, although the limited amount of rock recovered relative to the extent of this formation and its equivalents is a key consideration. These vents, charged with mantle CO_2 , would have had to have tapped into deep seated fluids leaching both the felsic basement and shallower mafic volcanics, as envisaged by Pietzsch *et al.* (2020).

In summary, the chemogenic BVF factory was unique amongst lacustrine deposystems in Earth history in terms of its scale and volume, but the Archean to Palaeoproterozoic also saw extensive precipitation of chemogenic carbonates in marine settings (Grotzinger & James, 2000). These have been attributed to high partial pressure of CO_2 and alkalinites in the oceans (Grotzinger & Kasting, 1993), and strong ocean stratification and upwelling triggering carbonate precipitation (Grotzinger & Knoll, 1995).

As illustrated in Fig. 1, the complex interactions of play of tectonics, relief, heat flow and volcanism likely active during the evolution of

lake basins, coupled with climatic controls, global atmospheric CO₂ levels and variations in the composition of the country rock, all likely interact to produce complex hydrological and hydrochemical scenarios. The BVF factory provides an example of such complexity, yet has a simple, if unique character more like the Archean than the Phanerozoic marine systems. This factory was the product of the complex process domain of a hyper-extended rifted margin, with a thinned crust and high heat flow, possibly in basins as much as 700 m or more below contemporaneous sea level. These basins had limited local run-off and siliciclastic input, and the ⁸⁷Sr/⁸⁶Sr ratios found in the carbonates suggest chemical sources from Cretaceous mafic volcanics and much older underlying basement but not contemporaneous oceans, in an endorheic system as discussed by Pietzsch *et al.* (2018, 2020). The BVF sediments reveal little about their depths of formation except for some wave-influenced grainstones–rudstones around structural highs. Hyper-alkaline lake waters, produced by high levels of CO₂, concentrated, at least in part, by evaporation, triggered the formation of Mg-silicate gels, whose presence or absence influenced the associated calcite textures, which indicate a predominance of abiotic (chemogenic) textures. Hyper-alkalinity was likely driven by high CO₂ levels, possibly directly into the lakes, but also as bicarbonate related to underlying and interbedded mafic-rich volcanics (Fig. 8). Carbon dioxide levels were also linked to global CO₂ events.

CONCLUSIONS

The vast Aptian hyper-alkaline lake systems of the central South Atlantic apparently have no analogues in the stratigraphic record and show more similarities with extensive chemogenic Archean–early Proterozoic marine carbonates than any known non-marine system. Many aspects remain controversial. There is still disagreement over how to interpret the seismic expression of these lake deposits, with some groups favouring the former presence of high relief, differentiated carbonate platforms rising many hundreds of metres above the lake basin floors. Others favour formation in shallow evaporitic lakes, with additional influences such as changes in chemoclines, or in the changed ratios of fluid sources. The chemogenic nature of the carbonates is still a contentious issue,

although the apparent paucity of microbial textures may reflect the high alkalinities and rapid precipitation rates evidenced by the mineral composition and textures. Another uncertainty relates to the formation of the Mg clays, which influenced carbonate textures, later diagenesis and porosity formation, because both chemogenic and microbially-influenced origins have been proposed. The problematic textures have required some researchers to propose new approaches to classification for these hybrid lithologies. The sources for the lake chemistry are unresolved with the likelihood that hydrothermal, CO₂-rich fluids released from the mantle, leached felsic basement and mafic volcanics, without any influence of marine waters or major fluvial influx. Global CO₂ levels likely also influenced these lakes, which themselves likely released mantle CO₂. The Barra Velha Formation (BVF) and its equivalents are an example of an extreme system, on a vast scale, due to a seemingly unique combination of mainly tectonically driven processes. Some carbonates are made, not born.

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